Fiscal Year:	FY 2022	**Task Last Updated:**	FY 08/30/2021
PI Name:	Mourkioti, Foteini Ph.D.		
Project Title:	Telomere Length Regulation in Muscle Atrophy		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Animal Biology: Vertebrate		
Space Biology Cross-Element Discipline:	(1) Musculoskeletal Biology		
Space Biology Special Category:	(1) Translational (Countermeasure) Potential		
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Comments:			
Project Type:	Ground,New Investigation	**Solicitation / Funding Source:**	2018 Space Biology (ROSBio) NNH18ZTT001N-FG. App B: Flight and Ground Space Biology Research
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No. of Post Docs:	1	**No. of PhD Degrees:**	3
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	2
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA ARC
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:	August 2021 report: No Key Personnel Changes		
COI Name (Institution):	Rompolas, Panteleimon Ph.D. (University of Pennsylvania)		
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(1) Introduction

During the NASA Twins Study, telomere length changes were observed in blood cells. Telomeres are specific sequences at the end of the chromosomes, protecting them from deterioration, much like the plastic tip at the end of a shoelace. Muscle atrophy is one of the main characteristics of astronauts in space. The only way to minimize muscle atrophy is through intensive exercise every day. However, the effects of telomere length changes in adult muscle cells remain unexplored. Interestingly, our lab recently developed a new method which allowed, for the first time, telomere length measurements in specialized cells within the adult muscles, including human tissues. We found that telomere shortening is a distinct feature of muscle stem cells (MuSCs) in atrophied muscles as well as in stem cells after repeated cycles of muscle damage and repair, similar to the ones observed in intensive exercise.

(2) Central objectives

Aim 1: Determine the combined effects of weightlessness and radiation exposure on telomere length of MuSCs.

Aim 2: Define the cellular outcomes MuSC-depended telomere length changes during muscle atrophy.

Aim 3: Investigate the molecular mechanism of telomere length in MuSCs.

Task Description:	(3) Methods to accomplish research objectives We will utilize the mouse hindlimb unloading model of disuse atrophy combined with gamma irradiation to evaluate their effect on telomere length, using a new method developed in our lab. Next, we will monitor for the first time in vivo the number of MuSCs using two-photon microscopy using genetically modified mice we recently generated and will further investigate the extent of apoptosis and senescence during atrophy. These studies are expected to provide a comprehensive analysis of the exact cellular dysfunction of shortened telomeres. Last, we will query the levels of a specific telomeric binding protein and its effect on persistent DNA damage at the telomeric sites. These experiments are expected to elucidate the molecular crosstalk leading to shortened telomeres in muscle cells. (4) Significance The goals of this research proposal align with NASA's emphasis on Animal biology (vertebrates) in Support of Human Space Exploration and the Cell and Molecular Biology guiding questions 1, 5, and 8 (to define the underlying mechanisms of the cells that are influenced by gravitational changes, causing tissue/organ dysfunction and affect stem cell function, respectively), as described at the Space Biology Science Plan 2016-2025. This work is highly relevant to the care of astronauts since we anticipate that our results will help understand the importance of telomere length alterations in stem cells during skeletal muscle atrophy that have not been studied before. Our study will benefit NASA and astronauts' health by 1) evaluating the combined effect of weightlessness and radiation exposure on telomere length in muscle atrophy and 2) assessing the molecular pathways affected during this process. Overall, by combining unique molecular and cellular biology approaches as well as genetically engineered mice that our lab has developed, we will determine the molecular mechanisms of telomere defects in MuSCs of atrophied muscles that can translate into benefits for NASA Exploration Mission. The regulatory mechanisms that we will explore in this proposal, using proven analogs such as mouse hindlimb unloading that induce biological effects similar to spaceflight, will be fundamental for future studies to investigate muscle specific telomere length changes in transitions from one gravity to another. Additionally, success in our experiments will inform similar mechanisms in musculoskeletal applications with stem cell dysfunction, such as muscle injuries after repeated exercises, providing benefits to the health and well being to both astronauts in space and those on Earth.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Success in our experiments will inform similar mechanisms in musculoskeletal applications with stem cell dysfunction, such as muscle injuries after repeated exercises, providing benefits to the health and well being to both astronauts in space and those on Earth.
Task Progress:	Previous efforts to measure telomere length in MuSCs had been hindered by the low abundance of stem cells within skeletal muscle, the heterogeneity of the muscle stem cell population, and the absence of an effective means of propagating pure MuSCs in the laboratory setting. Further, it was not previously known whether telomere attrition in diseased MuSCs contributed to a compromise in their regenerative potential, or whether MuSCs could self-renew following injury to give rise to functional cells. Using a protocol termed muscle quantitative fluorescence in situ hybridization, or MuQ-FISH, the Mourkioti Lab were able to quantify telomere length and number in prospectively isolated MuSCs and to establish that telomere attrition is present in human dystrophic MuSCs, underscoring their importance in diseased regenerative failure. The capacity to measure and quantify telomere length in MuSCs offers a practical tool to researchers developing measures to prevent muscle deterioration in space-like conditions. Previous studies have determined that substantial muscle atrophy (10% - 15% total mass) occurs in spaceflights of six months duration even among astronauts performing regular resistance exercise. Moreover, the recent NASA Twins Study determined a change in telomere length dynamics during spaceflight and within days of returning to Earth. Although skeletal muscle defects are a substantial issue for astronauts, the effects of telomere length changes in adult muscle cells has never been investigated previously. In this grant we measured telomere length alterations in muscle stem cells during skeletal muscle atrophy. In the coming year, we will continue our studies to determine the molecular mechanisms of telomere defects in MuSCs of atrophied muscles that can translate into benefits for NASA Exploration Mission.
Bibliography Type:	Description: (Last Updated: 01/09/2024)
Articles in Peer-reviewed Journals	Tichy ED, Ma N, Sidibe D, Loro E, Kocan J, Chen DZ, Khurana TS, Hasty P, Mourkioti F. "Persistent NF-kB activation in muscle stem cells induces proliferation-independent telomere shortening." Cell Rep. 2021 May 11;35(6):109098. https://doi.org/10.1016/j.celrep.2021.109098 ; PMID: 33979621; PMCID: PMC8183356 , May-2021
Articles in Peer-reviewed Journals	Tichy ED, Mourkioti F. "Telomere length assessments of muscle stem cells in rodent and human skeletal muscle sections." STAR Protocols. 2021 Dec 17;2(4):100830. Available online 14 September 2021. https://doi.org/10.1016/j.xpro.2021.100830 ; PMID: 34585163; PMCID: PMC8450295 , Dec-2021
Significant Media Coverage	Tichy ED, Ma N, Sidibe D, Loro E, Kocan J, Chen DZ, Khurana TS, Hasty P, Mourkioti F. "Research Summary of article 'Persistent NF-kB activation in muscle stem cells induces proliferation-independent telomere shortening' highlighted in Penn Medicine press release." Penn Medicine News press release, May 11, 2021. https://www.pennmedicine.org/news/news-releases/2021/may/pathway-behind-muscle-breakdowns-in-duchenne-muscular-dystrophy-discovered , May-2021